

Dheeraj Gajula

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EDUCATION

University of Colorado – Boulder

Boulder, CO

Master's in Computer Science

Aug 2025 – May 2027

- **Coursework:** Enterprise Networks, Network Systems, Machine Learning

Dayanada Sagar College of Engineering

Bangalore, India

Bachelor's in Computer Science and Engineering | 3.76/4.0

Nov 2020 – May 2024

- **Coursework:** Data structures and algorithms, Database systems, Operating systems, Computer Networks, Cloud Computing, Automata Theory, Software Engineering, Machine Learning, Deep Learning, NLP, Computer Vision

EXPERIENCE

Software Engineer – 1

June 2024 – Aug 2025

Versa Networks

Bangalore, India

- Developed REST APIs in **GoLang** and **Cassandra** that is serving more than 3000 reqs/s reported metrics through **Prometheus** and built dashboards through **Grafana**
- Performed **Quantitative and Qualitative analysis** of virus total malicious feed data by building multiple data pipelines using **Python** and **BigQuery** and built a **Mathematical Reinforcement model** to predict the result
- Containerized multiple services using **Docker** and **Kubernetes** and deployed them in **GCP**

Software Engineer – Intern

Feb 2024 – June 2024

Versa Networks

Bangalore, India

- Automated the device usage tracking at versa networks, reduced the time of billing from 7 days to under an hour
- Analyzed inconsistent logs, built systems to **detect anomalies**, and **StateMachines** to track device states
- Used **MongoDB** and **Python Data modelling** to process hierarchical data of the director logs and provided insights about the usage on **prometheus** and **Grafana**
- Used **Flask** and **FastAPI** for making it as a service, **Docker** and **Docker compose** for deploying it on servers

PROJECTS

Network Applications | *C++, network programming*

Sept 2025 – Present

- Developed a multi-threaded **TCP/UDP-based HTTP Web Server** in **C++** with request parsing for GET/HEAD/POST, proper status code handling, and a **UDP client/server file transfer system in C** supporting FTP-like commands (get, put, delete, ls, exit)
- Implemented **persistent connections with pipelining and timeout handling** in the HTTP server to serve multiple file types (.html, .txt, .png, .gif, .jpg, .ico, .css, .js) with correct Content-Type headers, and integrated **reliable transfer mechanisms** (Stop-and-Wait, Go-Back-N) to ensure robust UDP file transfers
- Built **concurrent request processing** using threading/forking for HTTP and acknowledgment-based communication for UDP, enabling multiple simultaneous client connections with graceful error handling and full protocol compliance

Enterprise Network Lab | *STP, DHCP Server, EIGRP, RIP, wireshark, wireless, VLANs*

Sept 2025 – Present

- Configured **Cisco enterprise networks** with VLANs, trunking (802.1Q/ISL), VTP, STP/RSTP, and **HSRP-based gateway redundancy** with uplink tracking for high availability
- Deployed **wireless LANs** with DHCP, multi-SSID VLAN segmentation, and routing/bridging on Cisco 1800 routers; implemented **NAT/PAT** (overload, static 1:1, pools) for Internet, DMZ, and VLAN address translation
- Implemented **routing protocols** (RIP v1/v2, EIGRP, OSPF multi-area) with redistribution, VRF isolation for HR traffic, DR/BDR elections, summarization, and optimized convergence across distributed Sales/R&D networks
- Performed **end-to-end validation** with Wireshark, extended ping/trace, SSH/Telnet, IOS recovery; verified NAT translations, OSPF adjacencies, VRF separation, and HSRP/SNAT failover persistence

TECHNICAL SKILLS

Languages: GoLang, Python, C/C++, SQL (Postgres), Bash

Frameworks and Database: Flask, FastAPI, Postgres, BigQuery, Cassandra, MongoDB, Prometheus, FireBase

Developer Tools: Git, Docker, Kubernetes, Google Cloud Platform, Grafana, AWS, NGINX, Hugging Face

Libraries: pandas, NumPy, Matplotlib, TensorFlow, OpenCV, scikit-learn, Seaborn, NLTK

ACHIEVEMENTS AND PUBLICATIONS

Explainable AI in the context of Schizophrenia

2024

- Presented our undergrad thesis at **International Conference for advanced data driven Intelligence**
- Built the project **Schizo AI**, explored different explainable AI models like **Lime**, **shap**, **CAM**

Smart India Hackathon winner

2022

- **Problem statement:** To come up with innovative solutions to save Court's time during the hearing day of a case
- Built project **Tenali**, **backend engineer** for the team

UNESCO India Africa Hackathon Finalist

2023

- **Problem Statement:** optimize the energy flow, Students from 22 countries attend this hackathon
- One of the 200 students that got selected from India, Received a Best Contributors award for the problem statement